IBM Quantum Hardware

Quick Recap

Quantum teleportation uses entanglement and classical communication to transfer a quantum state. Day 12 explains the physical systems that prepare, control and measure qubits when a circuit runs on real IBM hardware.

1. What Is IBM Quantum Hardware?

IBM Quantum Hardware is the complete set of physical devices used to create, control and measure superconducting qubits. It includes much more than the quantum processor chip.

- A Qiskit circuit passes through software and control hardware before reaching the qubits.
- Quantum operations are implemented using carefully calibrated microwave signals.
- Classical computers and the cloud deliver instructions and return measurement results.

Everyday analogy: A factory needs machinery, power, controllers and delivery systems—not just its main production machine.

2. Main Hardware Components

Component	What it does
Quantum Processing Unit (QPU)	Small chip containing superconducting qubits that perform quantum operations.
Dilution refrigerator	Keeps the processor extremely cold, around 15 millikelvin (0.015 K).
Microwave control electronics	Generate precise pulses to implement quantum gates.
Measurement electronics	Read microwave signals to identify measured 0 or 1 outcomes.
Classical computers and cloud	Coordinate jobs, process measurements and return results.

Figure 1. The quantum chip is only one component of the complete system.

3. Why Is Extreme Cooling Necessary?

Quantum states are sensitive to heat and environmental disturbances. IBM cools its superconducting processors close to absolute zero so their quantum behavior lasts long enough to perform operations.

- The dilution refrigerator reaches approximately 0.015 K.
- At very low temperatures, superconducting materials have almost zero electrical resistance.
- Cooling helps reduce disturbances and supports more reliable gates.

Everyday analogy: A quiet library makes it easier to hear a whisper; a controlled, very cold environment helps protect fragile qubit states. This analogy is not a physical explanation.

4. Superconducting Qubits and Quantum Gates

The QPU contains superconducting qubits on a chip. Gate instructions such as X, H and CNOT are converted into calibrated microwave pulse sequences that act on the qubits.

- Pulse frequency, duration, phase and amplitude must be controlled precisely.
- Two-qubit operations need coordinated control of connected qubits.
- Physical qubit connections are limited; not every pair can interact directly.

5. Qiskit Transpiler

The transpiler adapts an ideal quantum circuit to the selected processor’s physical layout and supported operations.

- Maps logical qubits to physical qubits.
- May add SWAP gates when qubits are not directly connected.
- Reorders and optimizes operations to reduce execution cost and errors.

Everyday analogy: A route planner adapts your preferred route to roads that actually exist.

6. From Python Code to Real Hardware

```
Python / Qiskit circuit
       |
Transpiler: map and optimize
       |
Control electronics: microwave pulses
       |
Cryogenic wiring -> QPU executes gates
       |
Measurement electronics read qubits
       |
Classical processing -> Cloud -> Python results
```

Figure 2. The hardware execution journey of a Qiskit circuit.

7. Measurement and Results

At the end of a circuit, measurement electronics detect small changes in microwave signals associated with qubit states. The readings are converted into classical bits and returned as counts or other measurement data.

- A measured qubit produces a classical 0 or 1.
- Repeated runs give distributions of outcomes.
- Noise and measurement imperfections can affect the counts.

8. Worked Example: A Two-Qubit Bell Circuit

```
from qiskit import QuantumCircuit
qc = QuantumCircuit(2)
qc.h(0)
qc.cx(0, 1)
qc.measure_all()
```

The Hadamard gate prepares superposition; CNOT creates entanglement; measurement produces two-bit results. Ideally, 00 and 11 appear about equally often. The workbook gives an illustrative noisy result of 00: 48%, 11: 46%, 01: 3%, 10: 3%.

9. Accessing IBM Hardware Through Qiskit

```
from qiskit_ibm_runtime import QiskitRuntimeService
service = QiskitRuntimeService()
backend = service.least_busy(simulator=False)
```

The workbook then introduces a Sampler job on the chosen backend. Actual execution requires a configured IBM Quantum account and an available processor; code interfaces can depend on installed versions.

10. Main Hardware Challenges

Challenge	Simple meaning
Decoherence	Qubits gradually lose quantum behavior through interaction with their environment.
Noise	Heat, vibration and electromagnetic disturbances alter qubit states.
Gate errors	Small imperfections accumulate as circuits become larger.
Measurement errors	Hardware may misread a 0 as 1 or a 1 as 0.

Current noisy systems are commonly described as NISQ (Noisy Intermediate-Scale Quantum) devices.

11. Classical vs Quantum Hardware

Classical computer	IBM quantum computer
Transistors and ordinary digital signals	Superconducting qubits and calibrated microwave pulses
Typically operates at room temperature	Requires extremely low operating temperature
Processes classical bits	Operates on quantum states; measurement returns bits

Classical computer	IBM quantum computer
General everyday computing	Specialized quantum computation and research

12. Applications Discussed

- Drug discovery: studying molecular interactions.
- Materials science: investigating batteries, superconductors and other materials.
- Finance: research into risk and portfolio optimization.
- Machine learning: hybrid AI and quantum research.
- Cryptography: understanding future quantum-safe security needs.
- Education: teaching circuits and developing algorithms.

13. Common Beginner Mistakes

- “The chip is the whole computer.” The cooling, controls, measurement systems and cloud also matter.
- “Quantum means a faster laptop.” Quantum processors target different classes of problems.
- “All qubits are connected.” Connectivity is limited and can require SWAP operations.
- “Real hardware gives perfect results.” Noise and errors can change measured outputs.

Summary: What to Remember

- IBM Quantum Hardware is an integrated hardware-and-software system.
- The QPU contains superconducting qubits.
- Dilution refrigerators cool the chip to about 15 mK.
- Microwave pulses physically implement quantum gates.
- The transpiler adapts circuits to the hardware layout.
- Measurement converts quantum outcomes into classical data.
- Real devices have decoherence, noise, gate errors and readout errors.
- Knowing hardware limitations helps you design better Qiskit circuits.